



Fig. 2: Safety indicators of DMMs

design for good grip while working, rugged body design with high-grade drop-proof material to ensure no break/damage even when dropped from a height, availability of warranty, service and affordable service parts are other important features to look for.

Standards and certifications

IEC 1010 standard is the safety standard for low-voltage test and measurement equipment. Look out for CE, UL, CSA and other symbols, safety categories (CAT certification) and EMC certification that would certify that the meter will work in inductive areas and so on. The DMM should have ingress protection (IP) certifications, which protect it against dust, water or any liquid and guarantees uninterrupted performance in harsh conditions. Look for IP ratings such as IP42, IP30, IP54, IP67, etc.

Calibration to standards of ISO or NIST is particularly important to companies that hold related certifications. These certifications indicate products that are made to very rigorous standards, so obtaining DMMs with correct certificates is important in these cases.

Wireless DMMs

Fluke 233 remote display DMM comes with an innovative wireless display and allows observation 9.1m (30-feet) away from the measurement point.

Fluke CNX 3000 wireless DMM is the captain of Fluke wireless team. It displays meter measurement plus readings from up to three wireless modules, as far as 20 metres away. If more than three readings are required from more modules, you can

add an optional PC adaptor to your laptop for real-time viewing of up to ten modules.

EXTECH EX540 and EX542 are twelve-function wireless true-rms industrial DMM/dataloggers (up to 9999 readings) with a wireless PC interface.

Smart DMMs

The new range of DMMs offers a new way to perform measurements much more accurately and comfortably with additional features like measurements through mobile applications. Using an app, the user can control any data remotely in real-time using a mobile phone or another mobile device with Android OS.

PROMAX PD-350, PD-351, PD-352 are DMMs with rms and Bluetooth control via an Android app. These also feature smart power-off function, which extends battery life.

OWON DMM - 35 category maintains features like voice warning supported to assure measurement safety, smart voice-reading, smart power-off option extending battery life, supports mobile device with Android 4.3 or above/iOS 7.0 or above and is equipped with BLE 4.0 module.

Keysight meter logger for Android smartphones and tablets improves productivity and safety by allowing the user to stream measurements in real-time from up to three Keysight DMMs and clamp meters wirelessly.

The future

For the most part, the future of DMMs will evolve slowly as these are well-establish, general-purpose test equipment that do not need to conform to rapidly-changing applications are. For such applications or those that need to address certain pa-

TABLE II SOME TOP DDMs	
Make/brand	Description
Amprobe (AM-560) 	- Amprobe AM-560 - Maximum AC/DC voltage: 1000V - Low-pass filter for VFD - Non-contact voltage detection enhances safety - CAT IV 600V and CAT III 1000V
Extech (EX330) 	- DC/AC voltage: 0.1mV to 600V - DC/AC current: 0.1µA to 10A - Temperature (type K): -4°F to 1382°F
Fluke (Fluke 117) 	- Fluke 117 electricians TRUE - Maximum voltage: 600V - Surge protection: 6kV peak - IEC 61010-1 600V CAT III - Display: Digital with 6000 counts, updates 4/sec - Battery life: 400 hours typical, without backlight
INNOVA (3320) 	- DC/AC voltage: 600V - DC current: 10A - AC current: 200mA - Large digital display 10 mega-ohm input impedance for safe automotive use - U/L listed for safety and quality
Mastech (MS8268) 	- DC voltage: 1000V - AC voltage: 750V - DC current: 10A - AC current: 10A - Display: 4000 counts - Display backlight - Auto power off 1000V CAT II and 600V CAT III
Triplet 1101-B 	- DC/AC voltage: Up to 600V - DC current: Up to 10A - Standard CAT II design - Battery life: 200 hours

rameters, there will be more specific meters that will address the needs of each application requirement, and the resulting instrument would not be called a DMM.

Where you will see some changes to DMMs in future will be inclusion of more memory for logging and graphing of data, along with wireless transmission of said data. **EFY**